Tutorial Week 3 – Intelligent Agents

Aim

- Review definitions
- Local python installation, Anaconda and accessing the aiama-python book code.
- Implement simple vacuum world agent.
- Work through TutorialWeek3.ipynb. Explore agent program for the different types of agents.

1. **Define in your own words the following terms: agent, percept, percept sequence, agent function, rational agent, task environment, properties of task environment.**
   Briefly describe the Agent Types discussed in the lecture and chapter 2 of the book.
   Discuss in class.

2. **Give a PEAS description of the task environment for a self driving car. What are the characteristics or dimensions of this task environment. See section 2.3 of the book.**

   Choose your own problem for an intelligent agent and give a PEAS description and characteristics of your chosen task environment. Discuss in class.

3. **Installing a local python environment and familiarization with aiama book code.**

We will primarily use Google Colab when working on our jupyter notebook files because it is more convenient. However, Colab has limitations on storage and processing time, which might make it harder to work with large codebases. Therefore, it is important to be comfortable with a local installation of python and setting up the relevant libraries. The sample code for the book will give you a good opportunity practice this. Work through the steps below to install an environment in Anaconda. Local installations of python environments will be similar.

Step 1. Find anaconda on Appsanywhere https://appsanywhere.westminster.ac.uk
Launch 'Jupyterlab and

Step 2. Open a terminal from within jupyterlab. You will see Terminal listed as an option.

Step 3. In your anaconda terminal type:

```
conda create -n py37 python=3.7
```

```
conda activate py37
```

Go back to environments in Anaconda Navigator and you should see the py37 environment you created. If jupyterlab does not work then try jupyter Notebook.

Step 4. Go to https://github.com/aimacode/aima-python and follow the instructions in the installation guide. If you do not have Git installed then download the zip file. Extract the file to your chosen drive. There should now be a directory called aima-python-master.

Step 5. Go back to Jupyter lab terminal and navigate to the aiama-python-mater folder and type:

```
pip install -r requirements.txt
```

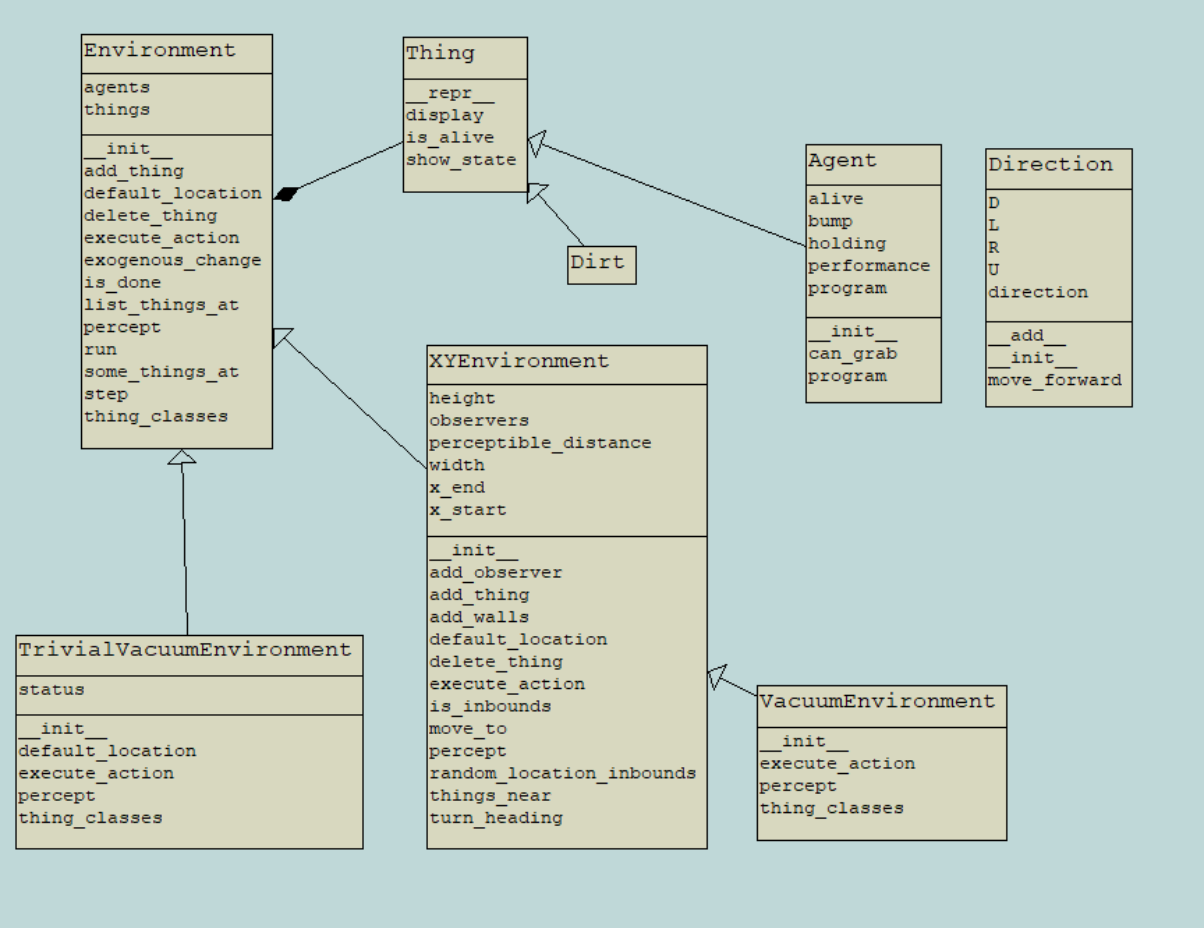
These commands will set up Python 3.7 and libraries needed to run the programs in the directory. (Be patient, this may take a while)

Step 6. Now you can examine the notebook ".ipynb" files in the aiama-python-master directory. Open and explore any ".ipynb" files you find interesting. These files will be helpful as examples when you start to work on your coursework.

4. Simple Vacuum world example.

Now, let's explore our first intelligent agent environment. We will recreate the simple vacuum agent example discussed in lecture 2.

Open the Jupiter notebook that implements the simple reflex agent described in the lecture notes. Refer to the TutorialWeek3.ipynb file. Feel free to use your Anaconda jupyter session from before, or Google Colab.
I've also created a class diagram to give you an overview of the structure of the agent implementation for your reference.



Familiarise yourself with the structure and code. Note that the intelligence in the agent is implemented in the program method of the Agent class. Examine each cell the nobebook and run it as you go along. Running the cells is essential to have the code work properly. Test yourself by attempting the questions in the notepad.

Explore more agent types such as TableDrivenVacuumAgent in the agents.ipynb file in the aiama github repository.

End.